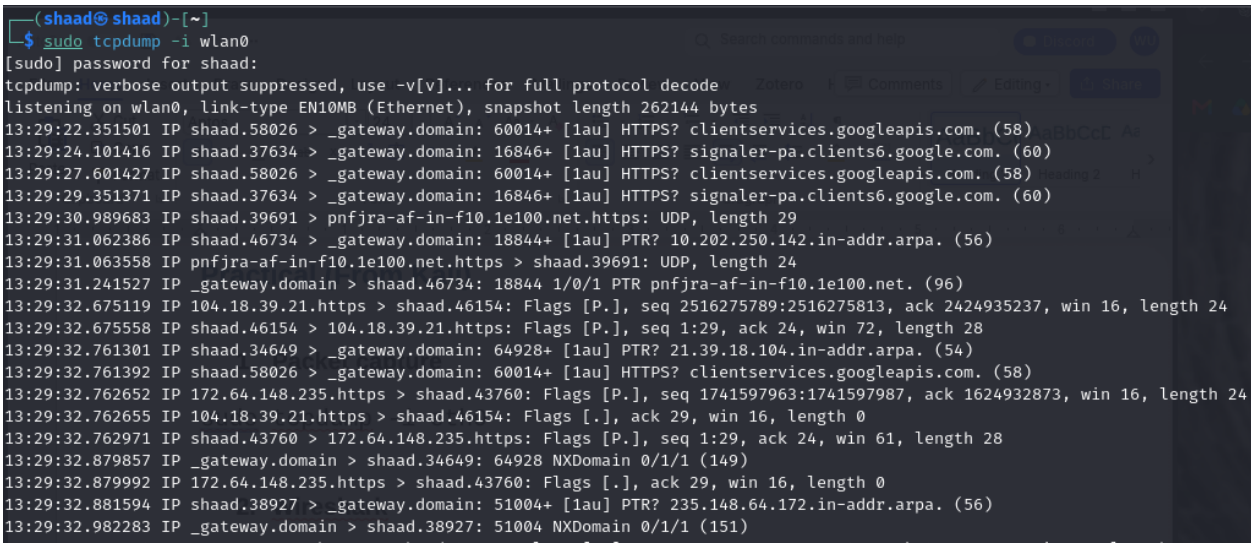
Day – 04 – Windows + networking

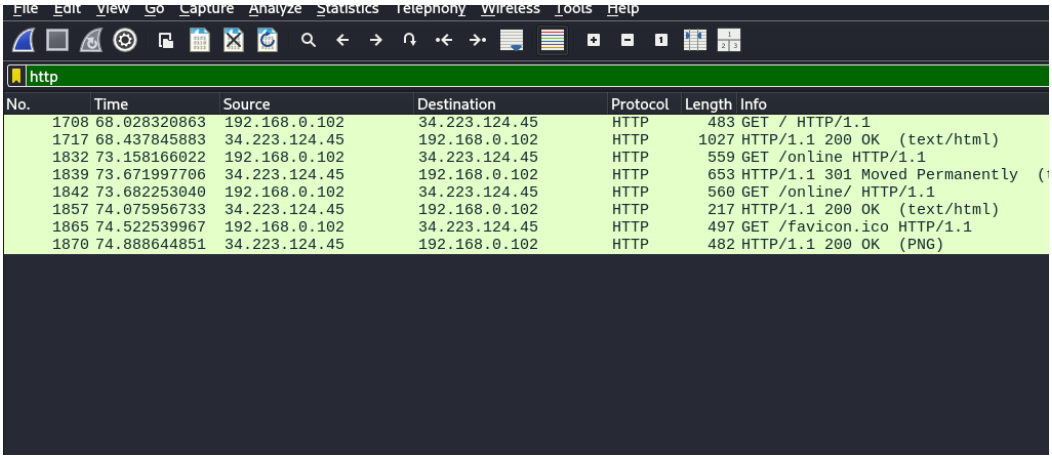
Packet Capture

Sudo tcpdump -i eth0 : Captures and displays network packets passing through the eth0 network interface

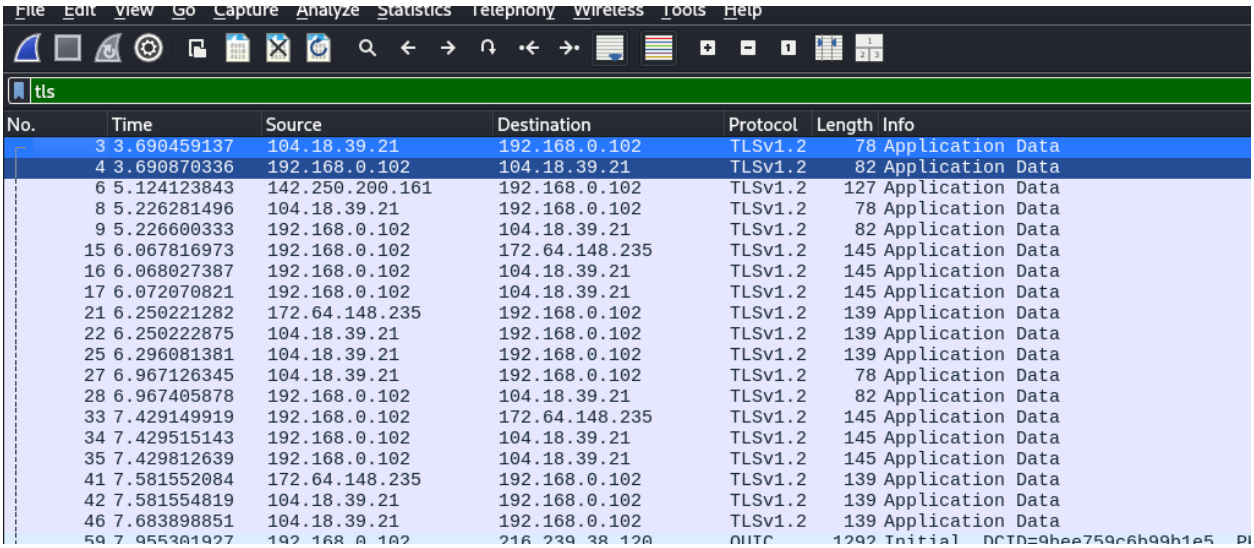


Wireshark

Http protocol:

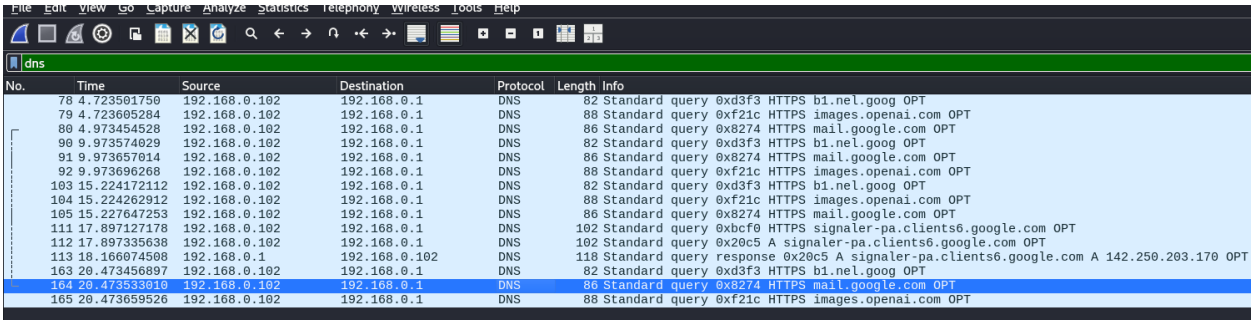


Https protocol:



No.	Time	Source	Destination	Protocol	Length	Info
3	3.690459137	104.18.39.21	192.168.0.102	TLSv1.2	78	Application Data
4	3.690870336	192.168.0.102	104.18.39.21	TLSv1.2	82	Application Data
6	5.124123843	142.250.200.161	192.168.0.102	TLSv1.2	127	Application Data
8	5.226281496	104.18.39.21	192.168.0.102	TLSv1.2	78	Application Data
9	5.226600333	192.168.0.102	104.18.39.21	TLSv1.2	82	Application Data
15	6.067816973	192.168.0.102	172.64.148.235	TLSv1.2	145	Application Data
16	6.068027387	192.168.0.102	104.18.39.21	TLSv1.2	145	Application Data
17	6.072070821	192.168.0.102	104.18.39.21	TLSv1.2	145	Application Data
21	6.250221282	172.64.148.235	192.168.0.102	TLSv1.2	139	Application Data
22	6.250222875	104.18.39.21	192.168.0.102	TLSv1.2	139	Application Data
25	6.296081381	104.18.39.21	192.168.0.102	TLSv1.2	139	Application Data
27	6.967126345	104.18.39.21	192.168.0.102	TLSv1.2	78	Application Data
28	6.967405878	192.168.0.102	104.18.39.21	TLSv1.2	82	Application Data
33	7.429149919	192.168.0.102	172.64.148.235	TLSv1.2	145	Application Data
34	7.429515143	192.168.0.102	104.18.39.21	TLSv1.2	145	Application Data
35	7.429812639	192.168.0.102	104.18.39.21	TLSv1.2	145	Application Data
41	7.581552084	172.64.148.235	192.168.0.102	TLSv1.2	139	Application Data
42	7.581554819	104.18.39.21	192.168.0.102	TLSv1.2	139	Application Data
46	7.683898851	104.18.39.21	192.168.0.102	TLSv1.2	139	Application Data
59	7.955301927	192.168.0.102	216.239.38.120	QUIC	1292	Initial, DCID=9bee759c6b99b1e5, PI

Observe DNS queries:

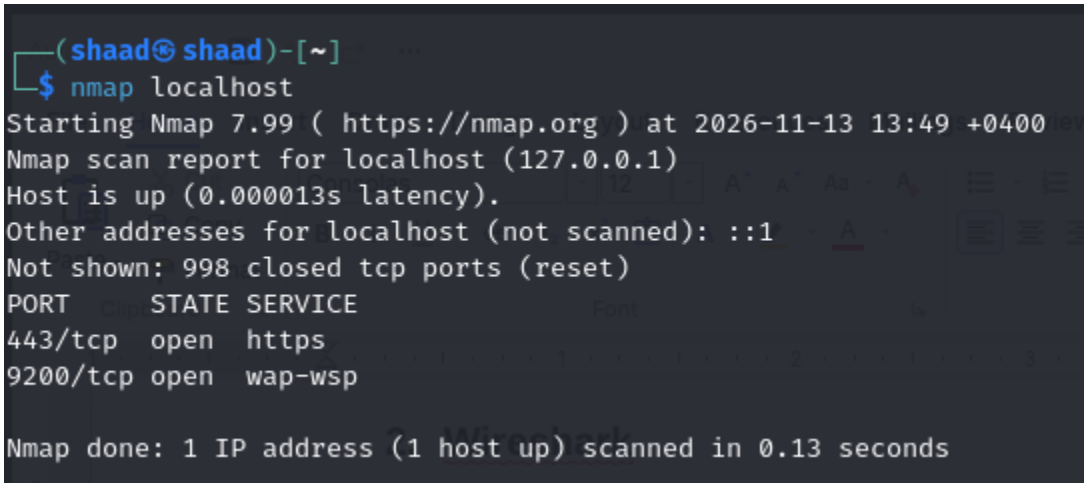


No.	Time	Source	Destination	Protocol	Length	Info
78	4.723501750	192.168.0.102	192.168.0.1	DNS	82	Standard query 0xd3f3 HTTPS b1.nel.goog OPT
79	4.723605284	192.168.0.102	192.168.0.1	DNS	88	Standard query 0xf21c HTTPS images.openai.com OPT
80	4.973454528	192.168.0.102	192.168.0.1	DNS	86	Standard query 0x8274 HTTPS mail.google.com OPT
90	9.973574029	192.168.0.102	192.168.0.1	DNS	82	Standard query 0xd3f3 HTTPS b1.nel.goog OPT
91	9.973657014	192.168.0.102	192.168.0.1	DNS	86	Standard query 0x8274 HTTPS mail.google.com OPT
92	9.973696268	192.168.0.102	192.168.0.1	DNS	88	Standard query 0xf21c HTTPS images.openai.com OPT
103	15.224172112	192.168.0.102	192.168.0.1	DNS	82	Standard query 0xd3f3 HTTPS b1.nel.goog OPT
104	15.224262912	192.168.0.102	192.168.0.1	DNS	88	Standard query 0xf21c HTTPS images.openai.com OPT
105	15.227647253	192.168.0.102	192.168.0.1	DNS	86	Standard query 0x8274 HTTPS mail.google.com OPT
111	17.897127178	192.168.0.102	192.168.0.1	DNS	102	Standard query 0xbcf0 HTTPS signaler-pa.clients6.google.com OPT
112	17.897335638	192.168.0.102	192.168.0.1	DNS	102	Standard query 0x20c5 A signaler-pa.clients6.google.com OPT
113	18.166074508	192.168.0.1	192.168.0.102	DNS	118	Standard query response 0x20c5 A signaler-pa.clients6.google.com A 142.250.203.170 OPT
163	20.473456897	192.168.0.102	192.168.0.1	DNS	82	Standard query 0xd3f3 HTTPS b1.nel.goog OPT
164	20.473533010	192.168.0.102	192.168.0.1	DNS	86	Standard query 0x8274 HTTPS mail.google.com OPT
165	20.473659526	192.168.0.102	192.168.0.1	DNS	88	Standard query 0xf21c HTTPS images.openai.com OPT

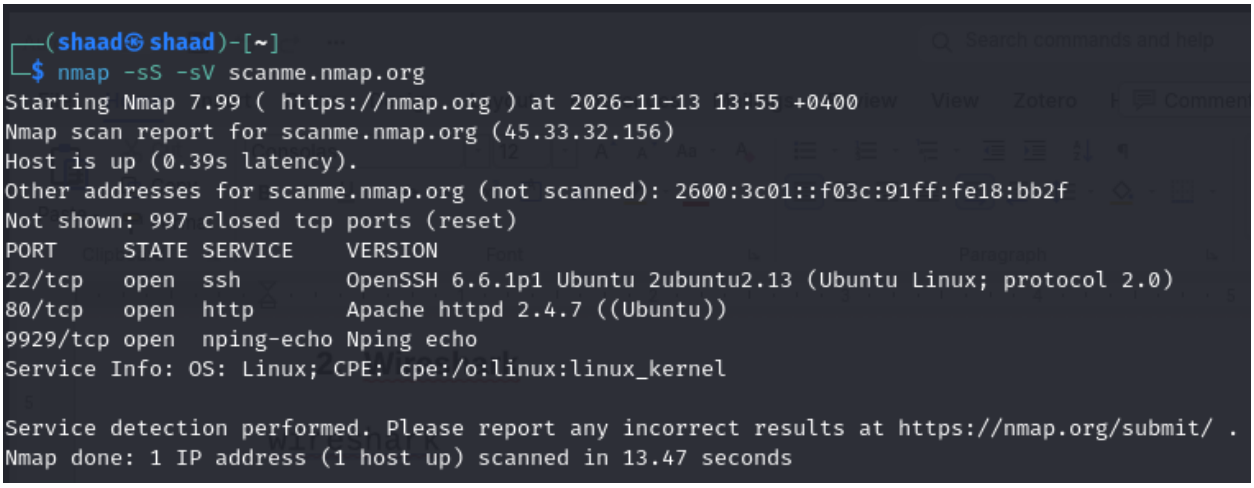
```
▸ Frame 164: Packet, 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface wlan0, id 0
▸ Ethernet II, Src: AzureWaveTec_2f:4a:7b (70:66:55:2f:4a:7b), Dst: DLinkInterna_cf:1b:fe (a0:a3:f0:cf:1
▸ Internet Protocol Version 4, Src: 192.168.0.102, Dst: 192.168.0.1
▸ User Datagram Protocol, Src Port: 52593, Dst Port: 53
▾ Domain Name System (query)
  ▾ Transaction ID: 0x8274
    ▸ [Expert Info (Warning/Protocol): DNS response missing]
  ▸ Flags: 0x0100 Standard query
    Questions: 1
    Answer RRs: 0
    Authority RRs: 0
    Additional RRs: 1
  ▾ Queries
    ▸ mail.google.com: type HTTPS, class IN
  ▸ Additional records
```

Service Enumeration

Nmap localhost : Scans local machine for open ports and identifies services running on them



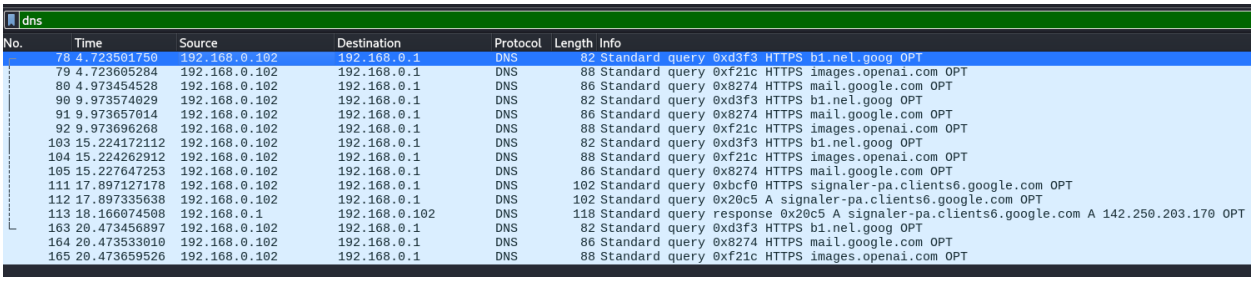
Nmap -sS -sV scanme.nmap.org : Performs a TCP SYN scan to identify open ports and detect the versions of services running on the target host.

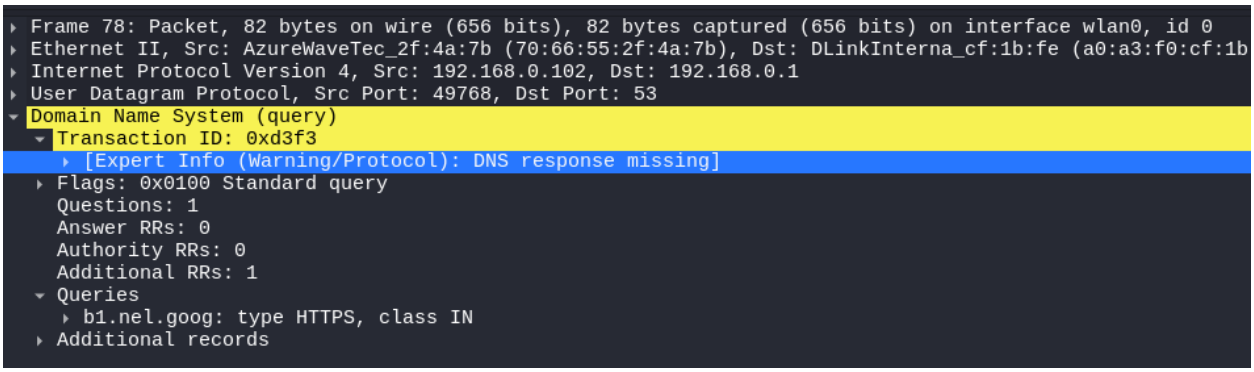


Hands on Task - Day-04

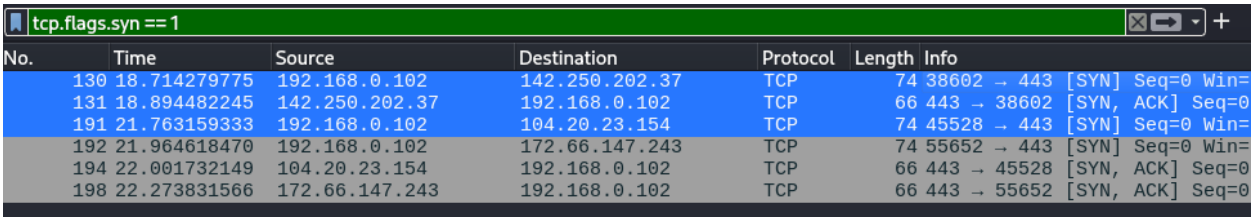
Capture traffic and identify:

DNS Request:





Tcp Handshake: first three shows three way handshake



Explain why HTTPS is secure:

HTTPS is secure because it uses TLS (Transport Layer Security) to encrypt communication between a browser and a web server. It protects sensitive information such as passwords and payment details from being intercepted or modified during transmission. It also verifies the server's identity using digital certificates, helping prevent impersonation and man in the middle attacks.